ITO Merchandising App

Infrastructure Cost Forecast

Assumptions

Parameter	Value	Notes
Daily image uploads	300 images	Shelf photos from store visits
Active users	6-8 users	Merchandisers in the field
Operating days	22 days/month	Business days only
Usage hours	Business hours	~10 hrs/day active usage
Image size (compressed)	~450 KB	JPEG @ 1280px max width
Monthly storage growth	~3 GB/month	300 x 22 x 450KB

Cloudflare R2 Storage Costs

Component	Rate	Year 1	Year 2	Year 3
Storage (cumulative)	$0.015/GB/mo	~36 GB	~72 GB	~108 GB
Storage cost	10 GB free	$0.39/mo avg	$0.93/mo avg	$1.47/mo avg
Class A ops (uploads)	$4.50/1M	Free tier	Free tier	Free tier
Class B ops (reads)	$0.36/1M	Free tier	Free tier	Free tier
Egress bandwidth	FREE	$0	$0	$0
Annual R2 Total		**$5**	**$11**	**$18**

* Free tier: 10GB storage, 1M Class A ops, 10M Class B ops/month

* Class A = uploads/writes (~6,600/mo); Class B = reads/views (~20,000/mo) - both well under free limits

Railway Compute + Database Costs

Component	Rate	Year 1	Year 2	Year 3
FastAPI compute	$0.028/vCPU-hr + $0.014/GB-hr	~$8/mo	~$10/mo	~$12/mo
PostgreSQL DB	Same rates + storage	~$10/mo	~$12/mo	~$15/mo
DB storage	$0.25/GB/mo	~$1/mo	~$2/mo	~$3/mo
Network egress	$0.10/GB	~$1/mo	~$1/mo	~$2/mo
Monthly Railway Total		**$20/mo**	**$25/mo**	**$32/mo**
Annual Railway Total		**$240**	**$300**	**$384**

* Containers run 24/7 even during off-hours (720 hrs/mo)

* Estimate: 0.25 vCPU avg, 512MB RAM for API; similar for PostgreSQL
* Growth assumes modest increase in data/traffic over time

3-Year Cost Summary

	Year 1	Year 2	Year 3	3-Year Total
Railway (Compute + DB)	$240	$300	$384	**$924**
Cloudflare R2 (Storage)	$5	$11	$18	**$34**
Annual Total	**$245**	**$311**	**$402**	**$958**
Monthly Average	$20/mo	$26/mo	$34/mo	**$27/mo avg**

Key Insights

> Railway (compute + DB) is 97% of costs; R2 storage is negligible due to free tier + no egress fees
> Costs grow ~30%/year as data accumulates and modest resource scaling
> R2 free tier covers operations; only pay for storage above 10GB
> Consider: Railway sleep/wake for off-hours could reduce compute by ~60%